vectors u and v. For one-dimensional vectors, this calculation is shown in Eq. (1).

$$cc(u,v) = \frac{\sum u_i v_i - \sum u_i \sum v_j}{\sqrt{\sum u_i^2 - (\sum u_i)^2}\sqrt{\sum v_j^2 - (\sum v_j)^2}} \tag{1}$$

The Euclidean distances between each two of those vectors are needed to process the HCA. Then there comes the need of Eq. (2) to calculate the distance d between the row/column vectors u and v.

$$d(u,v) = 1 - cc(u,v) = 1 - \frac{\sum u_i v_i - \sum u_i \sum v_j}{\sqrt{\sum u_i^2 - (\sum u_i)^2}\sqrt{\sum v_j^2 - (\sum v_j)^2}} \tag{2}$$

Briefly speaking, the correlation matrix is calculated firstly, with which the distance matrix can be calculated and used for the HCA process. It is clear that HCA function needs to perform matrix calculations, so there are mathematically limitations. If there are any blank items in the data, or if the data matrix is a singular matrix, HCA won't be possible.

At the code level, this program uses SciPy to complete the calculation process (Oliphant, 2007). This article only describes the main idea instead of the specific calculation process, which can be seen in code, at https://github.com/GeoPyTool/GeoPyTool/blob/master/geopytool/Cluster.py .

Specifically, take Table 10 as an example, in which horizontal rows and vertical columns are respectively considered as comparative indexes. Each row is considered as a one-dimensional vector. Euclidean distances between each two rows is used for the clustering of left dendrogram shown in Fig. 2e. Similar process for each column generates the dendrogram in the upper part of Fig. 2e. The left dendrogram and the clustering for horizontal rows can be used to separate ungrouped data into several groups, or to estimate the dissimilarity of different rows. The upper dendrogram and the clustering for vertical columns can be used to evaluate correlation between different oxides or elements. The colored matrix in the middle of Fig. 2e is a combination of the correlation matrix of rows and columns.

Data in Table 10 contain different samples of a same volcanic rock. It can be seen from Fig. 2e that these samples can be divided into two groups by HCA, and SiO_2 and other oxides is significantly dissimilar, which hints that there may be other sources of ingredients that affect the silica content.

5.4.3. Developing functions

Besides the routines mentioned above, new functions have been added, including a clay-silt-sand classification diagram (Fig. 2f and data in Table 11), a 3D visualization function (Fig. 2g), and an Auto function to generate all available calculations and plots for imported data. GeoPyTool is still in development, more functions will be gradually added.

5.5. Output files

GeoPyTool generates MS Excel XLSX and CSV files as output files containing calculation result and generates diagrams in a PNG, SVG or PDF format. The PNG format has become an International Standard (ISO/IEC 15948:2003) and was classified as a World Wide Web Consortium (W3C) recommendation in 2003 (Suyanto, 2008; Web, 2010). This format is also supported by all mainstream operating systems and provides images with a decent quality. The SVG format is a widely deployed royalty-free graphic format developed and maintained by the W3C SVG Working Group. It is supported by all modern browsers for both desktop and mobile computers (Quint, 2003). The PDF format is also widely used as a universal format for documents and is used in GeoPyTool as a complementary and extra format to SVG. Major graphic programs

Table 11
Sand-silt-clay data (vol.%) sample for GeoPyTool.

Label	Marker	Color	Size	Alpha	Style	Width	Clay	Sand	Silt
First Group	o	green	50	0.6	-	0.4	60.0	25.0	15.0
	o	green	50	0.6	-	0.4	44.0	31.0	25.0
	o	green	50	0.6	-	0.4	62.6	18.7	18.7
	o	green	50	0.6	-	0.4	67.7	9.0	23.3
	o	green	50	0.6	-	0.4	66.6	21.8	11.6
	o	green	50	0.6	-	0.4	67.5	18.8	13.7
	o	green	50	0.6	-	0.4	59.5	20.5	20
	o	green	50	0.6	-	0.4	69.8	15.2	15
	o	green	50	0.6	--	0.5	59.8	15.2	25
	o	green	50	0.6	--	0.5	62.6	18.7	18.7
	o	green	50	0.6	--	0.5	61.3	16.8	21.9
	o	green	50	0.6	--	0.5	58.3	20	21.7
	o	green	50	0.6	--	0.5	66.7	13.0	20.3
	o	green	50	0.6	--	0.5	51.5	30.5	18.0
	o	green	50	0.6	--	1.5	74.3	11.4	14.3
Second Group	d	blue	50	0.6	--	2.5	50.0	35.0	15.0
	d	blue	50	0.6	--	3.5	54.4	18.4	27.2
	d	blue	50	0.6	--	4.5	80.3	11.8	7.9
	d	blue	50	0.6	--	5.5	51.3	27.1	21.6
	d	blue	50	0.6	--	6.5	56.4	28.3	15.3
Third Group	D	grey	50	0.6	--	7.5	58.3	20.0	21.7
	D	grey	50	0.6	--	8.5	66.7	18.0	15.3
	D	grey	50	0.6	--	0.5	76.0	12.0	12.0
	D	grey	50	0.6	--	0.5	71.9	13.5	14.6
	D	grey	50	0.6	--	0.5	70.0	17.5	12.5
	D	grey	50	0.6	--	0.5	69.8	11.0	19.2
The Last Group	s	red	50	0.6	--	0.5	51.8	48.2	0.0
	s	red	50	0.6	--	0.5	40.8	59.2	0.0
	s	red	50	0.6	--	1.5	61.0	39.0	0.0
	s	red	50	0.6	--	2.5	40.0	60.0	0.0